MIL-G-50250 (MU)
15 February 1969
(SEE 6.6)

MILITARY SPECIFICATION

GRENADE, HAND, PRACTICE, DELAY, M62 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing of the M62 Practice Hand Grenade.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1330

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9231597 - Grenade, Hand, Practice, Delay, M62
7548339 - Container, Fiber, Ammunition, M289A2
7548340 - Box, Packing, Ammunition

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 9231597 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Grenade assembly.- The grenade assembly shall comply with all requirements specified on Drawing (dwg.) 9231597 and with all requirements specified in applicable specifications.

3.3 Functioning.- The grenade assembly shall emit a cloud of smoke and a loud report, and shall show no evidence of rupture or separation of the metal parts, when tested as specified in 4.4.1.

3.4 Moisture content of black powder.- The moisture content of the black powder at the loading station at the time of loading shall not exceed 0.5 percent, when tested as specified in 4.4.2.

3.5 Release resistance.- The safety clip shall not be disengaged from its assembled position over the lever by a load of not less than twenty-five (25) ounces when tested as specified in 4.4.3. (see 6.5).

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3.6 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, chips, scale, dirt, grease, rust, salt deposits and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.7 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

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c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a sample of thirty (30) loaded assemblies to the Government for testing in accordance with the provisions of 4.2.2. All parts and materials shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 Inspections to be performed.- Assemblies will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification. The functioning test shall be performed in accordance with 4.4.1.

4.2.3 Rejection.- If any assembly fails to comply with any or all of the specified requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly in the sample to comply with any or all of the requirements.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable

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requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded practice hand grenades shall contain:

a. Parts of one lot interfix number from one manufacturer.

b. Black powder from not more than one lot from one manufacturer.

c. M205A2 practice grenade fuzes from not more than two serial lots of one lot interfix from one manufacturer. These lots shall be consecutively produced lots or if the adjacent fuze serial lot is not available, the second fuze lot shall not vary from the first fuze lot by more than three serial lot numbers.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major

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and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Charge assembly (see dwg. 8861650 covering a detail of dwg. 9231597).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Weight of charge.....	Balance	01001
102.	Bag torn or perforated as evidenced by leaking of powder charge.....	Visual	01002
103.	Heat seal missing or incomplete.	Visual-Manual	01003
Minor:	AQL 1.50 percent		
201.	Width of seal.....	Gage	01004
202.	Length of bag.....	Gage	01005
203.	Evidence of poor workmanship (see 3.6).....	Visual	01006

4.3.2.2 Assembly, prior to assembling fuze (see dwg. 9231597).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Charge assembly missing.....	Visual	02001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	02002

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4.3.2.3 Assembly (see dwg. 9231597).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Safety pin not properly spread..	Visual	03001
2.	Safety clip missing or incorrectly assembled.....	Visual	03002
Major:	AQL 0.40 percent		
101.	Incorrect fuze.....	Visual	03003
102.	Component other than safety clip missing.....	Visual	03004
Minor:	AQL 1.50 percent		
201.	Fuze loose.....	Manual	03005
202.	Protective coating contains bare spot.....	Visual	03006
203.	Marking misleading or unidentifiable, including incorrect color code.....	Visual	03007
204.	Evidence of poor workmanship (see 3.6).....	Visual	03008

4.3.2.4 Sealed container (see dwg. 7548339).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Tape incomplete or badly wrinkled.....	Visual	04001
102.	Metal end loose or distorted....	Visual-Manual	04002
103.	Cut or damage through all layers.....	Visual-Manual	04003

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 2.50 percent		
201.	Length of tear tab, minimum (min.).....	Scale	04004
202.	Color of tape incorrect.....	Visual	04005
203.	Marking misleading or unidentifiable.....	Visual	04006
204.	Contents loose.....	Manual	04007
205.	Excessive gap between cover and body.....	Visual	04008

4.3.2.5 Sealed packing box (see dwg. 7548340).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Board broken to extent that contents are exposed or liable to become exposed.....	Visual	05001
102.	DOD symbol misleading or unidentifiable.....	Visual	05002
103.	Hardware missing, broken or loose.....	Visual-Manual	05003
104.	Strapping missing, broken or loose.....	Visual-Manual	05004
Minor:	AQL 2.50 percent		
201.	Marking misleading or unidentifiable.....	Visual	05005
202.	Hardware or strapping improperly engaged.....	Visual-Manual	05006
203.	Car seal missing, unsealed or improperly positioned.....	Visual-Manual	05007
204.	Handle missing or insecure.....	Visual-Manual	05008
205.	Contents loose.....	Manual	05009

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4.3.3 Production testing.- The following tests shall be performed using MIL-STD-105 for sampling plan requirements unless otherwise specified.

4.3.3.1 Functioning (see 3.3) - Major defect - Code No. 03009

4.3.3.1.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the specified acceptance criteria, one hundred and twenty-five (125) assemblies shall be selected from each lot for test. The lot shall be rejected if three (3) or more assemblies fail to comply with the requirements. The test shall be performed as specified in 4.4.1.

4.3.3.1.2 After three (3) consecutive lots have complied with the criteria of 4.3.3.1.1, the sampling plan shall be in accordance with Code Letter H, and an AQL of 1.00 percent.

4.3.3.2 Moisture content of black powder (see 3.4) - Major defect - Code No. 06001.- The contractor shall provide adequate controls to insure that the powder complies with the requirements. For verification, the contractor shall select and test one sample of black powder from each eight hours of production of charge assemblies. A composite sample shall not be used. If the moisture content of a sample exceeds the requirement, all the non-conforming material shall be rejected including any charge assemblies containing the non-conforming material. The test procedure shall be in accordance with 4.4.2.

4.3.3.3 Release resistance (see 3.5) - Major defect - Code No. 03010.- The sampling plan for this test shall be in accordance with MIL-STD-1235, CSP-1, with an AQL of 0.10 percent, wherein I = 800 and f = 1/50.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 9231597 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

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4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Functioning.- The grenade assembly less safety clip shall be placed in a holding fixture and fired by removing the safety pin. The grenade assembly shall be observed for compliance with the requirements.

4.4.2 Moisture content of black powder.- The moisture content of the black powder shall be determined as specified in MIL-P-223.

4.4.3 Release resistance.- The load shall be applied slowly and steadily parallel with the safety pin in the clip releasing direction until the total load is supported by the clip. The clip shall be observed for compliance with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.- The grenade assemblies shall be preserved and packaged in accordance with dwg. 7548339.

5.2 Packing

5.2.1 Level A.- The fiber containers shall be packed in accordance with dwg. 7548340.

5.3 Marking.- Marking shall be as specified on dwgs. 7548339 and 7548340.

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6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

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CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of load test data and ammunition data cards.- In addition to the normal distribution of records, one copy of all loaded test data and ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SMUPA-ND5
(2) SMUPA-DC.

6.5 The safety clip shall also be considered disengaged from its assembled position over the lever when the square end of the safety clip moves upon the flat surface of the lever to the extent that the clip does not return to its original assembled position when the load is removed.

6.6 Supersession data.- This specification includes the requirements of PA-PD-3081, dated 15 January 1968.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1330-A-203

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

PICATINNY ARSENAL
DOVER, N.J.
ATTN: SMUPA-ND

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-G 50250(MU)

ORGANIZATIC *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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